

Another useful consequence of the intermediate value theorem is the following. The interval referred to may be either closed, open, half-open, or infinite.

Theorem (2.27)

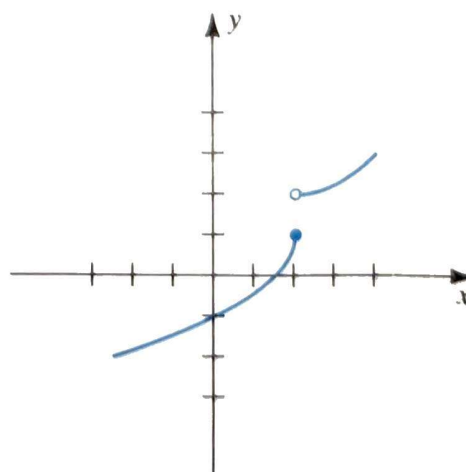
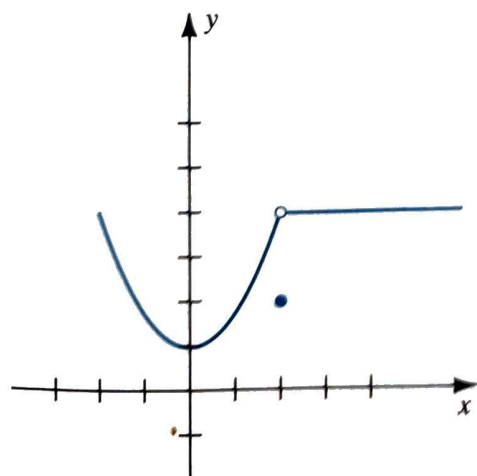
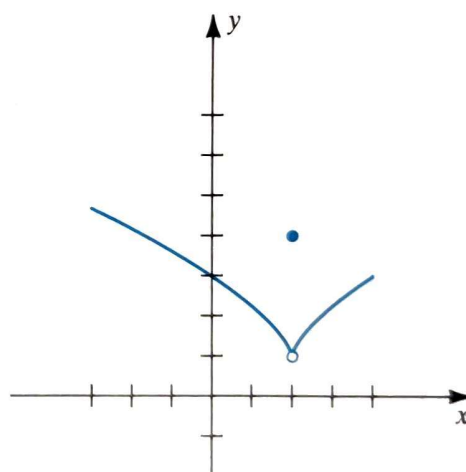
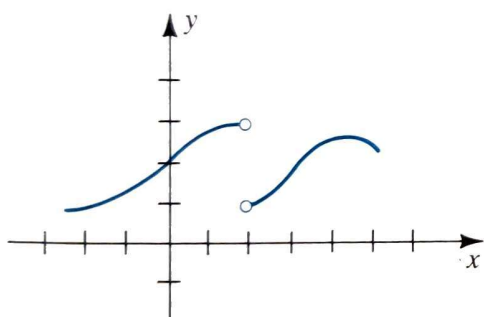
If a function f is continuous and has no zeros on an interval, then either $f(x) > 0$ or $f(x) < 0$ for every x in the interval.

PROOF The conclusion of the theorem states that under the given hypothesis, $f(x)$ has the same sign throughout the interval. If this conclusion were *false*, then there would exist numbers x_1 and x_2 in the interval such that $f(x_1) > 0$ and $f(x_2) < 0$. By our preceding remarks, this, in turn, would imply that $f(c) = 0$ for some number c between x_1 and x_2 , contrary to hypothesis. Thus, the conclusion must be true. ■

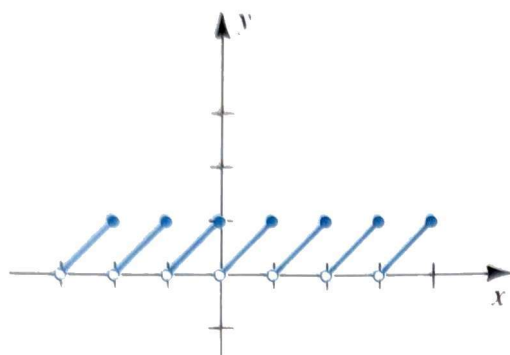
In Chapter 4 we shall apply Theorem (2.27) to the derivative of a function f to help determine the manner in which $f(x)$ varies on various intervals.

EXERCISES 2.5

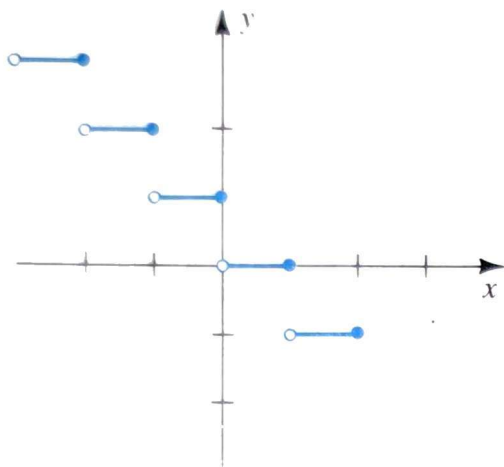
Exer. 1–10: The graph of a function f is given. Classify the discontinuities of f as removable, jump, or infinite.



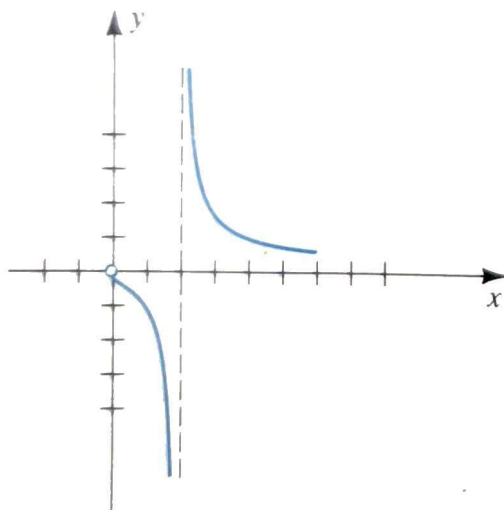
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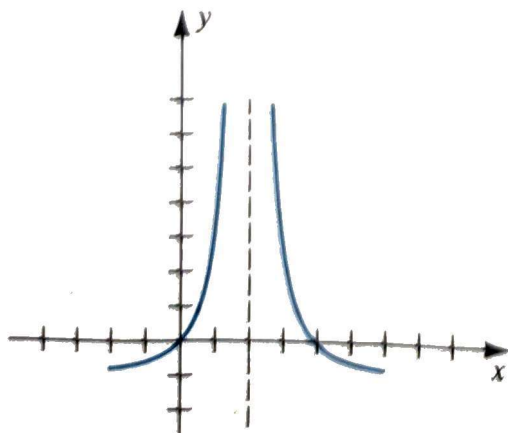
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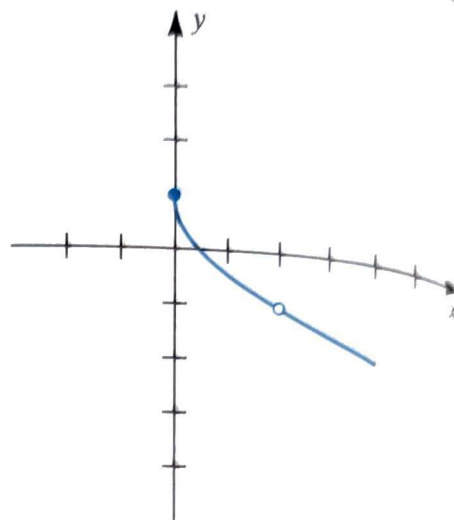
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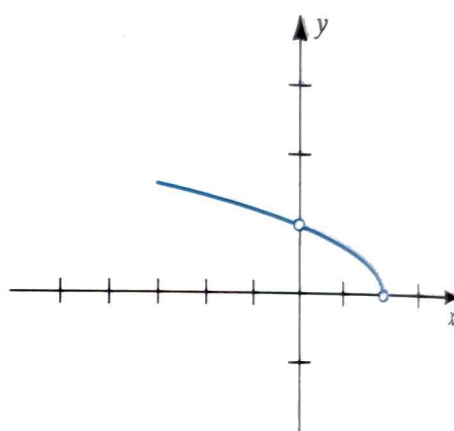
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Exer. 11–18: Classify the discontinuities of f as removable, jump, or infinite.

$$11 \quad f(x) = \begin{cases} x^2 - 1 & \text{if } x < 1 \\ 4 - x & \text{if } x \geq 1 \end{cases}$$

$$12 \quad f(x) = \begin{cases} x^3 & \text{if } x \leq 1 \\ 3 - x & \text{if } x > 1 \end{cases}$$

$$13 \quad f(x) = \begin{cases} |x + 3| & \text{if } x \neq -2 \\ 2 & \text{if } x = -2 \end{cases}$$

$$14 \quad f(x) = \begin{cases} |x - 1| & \text{if } x \neq 1 \\ 1 & \text{if } x = 1 \end{cases}$$

$$15 \quad f(x) = \begin{cases} x^2 + 1 & \text{if } x < 1 \\ 1 & \text{if } x = 1 \\ x + 1 & \text{if } x > 1 \end{cases}$$

$$16 \quad f(x) = \begin{cases} -x^2 & \text{if } x < 1 \\ 2 & \text{if } x = 1 \\ x - 2 & \text{if } x > 1 \end{cases}$$

$$[c] \quad 17 \quad f(x) = x^{-1/3} \sin \left[\cos \left(\frac{\pi}{2} - x^2 \right) \right]$$

$$[c] \quad 18 \quad f(x) = \frac{\sin(x^2 - 1)}{(x - 1)^2}$$

Exer. 19–22: Show that f is continuous at a .

19 $f(x) = \sqrt{2x-5} + 3x$; $a = 4$

20 $f(x) = \sqrt[3]{x^2+2}$; $a = -5$

21 $f(x) = 3x^2 + 7 - \frac{1}{\sqrt{-x}}$; $a = -2$

22 $f(x) = \frac{\sqrt{x}}{2x+1}$; $a = 8$

Exer. 23–30: Explain why f is not continuous at a .

23 $f(x) = \frac{3}{x+2}$; $a = -2$

24 $f(x) = \frac{1}{x-1}$; $a = 1$

25 $f(x) = \begin{cases} \frac{x^2-9}{x-3} & \text{if } x \neq 3 \\ 4 & \text{if } x = 3 \end{cases}$ $a = 3$

26 $f(x) = \begin{cases} \frac{x^2-9}{x+3} & \text{if } x \neq -3 \\ 2 & \text{if } x = -3 \end{cases}$ $a = -3$

27 $f(x) = \begin{cases} 1 & \text{if } x \neq 3 \\ 0 & \text{if } x = 3 \end{cases}$ $a = 3$

28 $f(x) = \begin{cases} \frac{|x-3|}{x-3} & \text{if } x \neq 3 \\ 1 & \text{if } x = 3 \end{cases}$ $a = 3$

29 $f(x) = \begin{cases} \frac{\sin x}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$ $a = 0$

30 $f(x) = \begin{cases} \frac{1-\cos x}{x} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}$ $a = 0$

Exer. 31–34: Find all numbers at which f is discontinuous.

31 $f(x) = \frac{3}{x^2+x-6}$

32 $f(x) = \frac{5}{x^2-4x-12}$

33 $f(x) = \frac{x-1}{x^2+x-2}$

34 $f(x) = \frac{x-4}{x^2-x-12}$

Exer. 35–38: Show that f is continuous on the given interval.

35 $f(x) = \sqrt{x-4}$; $[4, 8]$

36 $f(x) = \sqrt{16-x}$; $(-\infty, 16]$

37 $f(x) = \frac{1}{x^2}$; $(0, \infty)$

38 $f(x) = \frac{1}{x-1}$; $(1, 3)$

Exer. 39–54: Find all numbers at which f is continuous.

39 $f(x) = \frac{3x-5}{2x^2-x-3}$

40 $f(x) = \frac{x^2-9}{x-3}$

41 $f(x) = \sqrt{2x-3} + x^2$

42 $f(x) = \frac{x}{\sqrt[3]{x-4}}$

43 $f(x) = \frac{x-1}{\sqrt{x^2-1}}$

44 $f(x) = \frac{x}{\sqrt{1-x^2}}$

45 $f(x) = \frac{|x+9|}{x+9}$

46 $f(x) = \frac{x}{x^2+1}$

47 $f(x) = \frac{5}{x^3-x^2}$

48 $f(x) = \frac{4x-7}{(x+3)(x^2+2x-8)}$

49 $f(x) = \frac{\sqrt{x^2-9}\sqrt{25-x^2}}{x-4}$

50 $f(x) = \frac{\sqrt{9-x}}{\sqrt{x-6}}$

51 $f(x) = \tan 2x$

52 $f(x) = \cot \frac{1}{3}x$

53 $f(x) = \csc \frac{1}{2}x$

54 $f(x) = \sec 3x$

Exer. 55–58: Verify the intermediate value theorem (2.26) for f on the stated interval $[a, b]$ by showing that if $f(a) \leq w \leq f(b)$, then $f(c) = w$ for some c in $[a, b]$.

55 $f(x) = x^3 + 1$; $[-1, 2]$

56 $f(x) = -x^3$; $[0, 2]$

57 $f(x) = x^2 - x$; $[1, 3]$

58 $f(x) = 2x - x^2$; $[-2, -1]$

59 If $f(x) = x^3 - 5x^2 + 7x - 9$, use the intermediate value theorem (2.26) to prove that there is a real number a such that $f(a) = 100$.

60 Prove that the equation $x^5 - 3x^4 - 2x^3 - x + 1 = 0$ has a solution between 0 and 1.

61 In models for free fall, it is usually assumed that the gravitational acceleration g is the constant 9.8 m/sec² (or 32 ft/sec²). Actually, g varies with latitude. If θ is the latitude (in degrees), then a formula that approximates g is

$$g(\theta) = 9.78049(1 + 0.005264 \sin^2 \theta + 0.000024 \sin^4 \theta).$$

Use the intermediate value theorem to show that $g = 9.8$ somewhere between latitudes 35° and 40°.